Supplemental Material for: Why sodium cobaltate superconducts only when it is wet

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This document is the engine room of the Letter. It carries the data, the derivations, and the checks that the four-page format could only summarize, in the same order the Letter argues them: methods, the well transition, the coupling model, the electronic evidence for the gallery band, the water, the magnetism, the pairing state, the correlations we have not treated, the design rule and its one counterexample, and the predictions that reach beyond this one compound.

S1. METHODS

Every number quoted in the Letter comes from one of the calculation sets described here, in enough detail that any of them can be checked or repeated without consulting the code. Table S1 gives the inventory; the rest of this section gives the settings.

Code and functional. All total energies are spin-polarized density-functional calculations in the PBE generalized-gradient approximation, using projector-augmented-wave (PAW) pseudopotentials from pslibrary, run in Quantum ESPRESSO 7.3.1 (GPU build). Plane-wave cutoffs are 60 Ry (wavefunctions) and 480 Ry (density) throughout. Electronic self-consistency uses Marzari–Vanderbilt cold smearing of 0.02 Ry, local Thomas–Fermi charge mixing with mixing_beta = 0.3, and a convergence threshold conv_thr = 10^{-7} Ry for all production total-energy scans; the single exception is the Born–Oppenheimer molecular-dynamics stage (set L, Sec. S7), where conv_thr is loosened to 10^{-6} Ry purely for throughput and every other setting matches set G. Every calculation is spin-polarized (nspin = 2); a direct nspin = 1 vs. nspin = 2 check is reported below, and the local-spin-density over-polarization that separately affects the $\sqrt{3} \times \sqrt{3}$ magnetic moments is discussed in Sec. S10.

Cells. The full-coverage 1×1 cell ($x = 1$, one alkali per formula unit) is used for the main displacement scans, in-plane lattice constants $a = 2.888$ Å (Na), 2.82 Å (Li), and 2.90 Å (K; an ansatz upper estimate bracketing the experimental 2.84–2.87 Å range for K_xCoO_2), with cell areas 7.22, 6.89, and 7.28 Å^2 respectively. The physical $\sqrt{3} \times \sqrt{3}$ cell ($x = 1/3$, $a = 5.00$ Å, area 21.67 Å^2) checks that the transition survives at the real sodium content. Four CoO_2–CoO_2 spacings are computed for each series, $c = 5.5$, 6.9, 8.4, 9.9 Å, bracketing the anhydrous, monolayer-hydrate, and bilayer-hydrate structures actually reported. The alkali sits on the Co-top in-plane site (the lowest-energy registry found in the earlier surface work [1]) at $z = c/2 + \delta$; the oxygen height above the Co plane is held fixed at $z_O = 0.96$ Å for every spacing and every δ, the largest systematic in the calculation (below).

k-meshes. $8 \times 8 \times 3$ for the 1×1 displacement scans, $6 \times 6 \times 3$ for the $\sqrt{3} \times \sqrt{3}$ scans, $6 \times 6 \times 2$ (Gamma-centered) for the explicit-water cell (set G/J/L, Sec. S7), and a dense $24 \times 24 \times 8$ mesh for the non-self-consistent eigenvalues that feed $N(0)$, the density of states, and the Fermi-surface maps of Sec. S4. The band-structure/fatband runs of Sec. S5 follow a 60-point Γ–M–K–Γ path with PROJWFC.X projections onto all Na s/p and Co d orbitals.

Fitting and quantization. $E(\delta)$ is fit to the even Landau polynomial $E_0 + \alpha\delta^2 + \beta\delta^4$ (Eq. (S1)); where the quartic fit is not bracketed by the sampled δ range (the $c \geq 8.4$ Å series, Sec. S2), it is replaced by a constrained sextic, $E_0 + \alpha\delta^2 + \beta\delta^4 + \gamma_6\delta^6$ with $\gamma_6 > 0$ enforced and a hard-core cap $\delta_{\min} \leq c/2 - 2.0$ Å (an ansatz for the alkali–oxygen contact distance; varying it 1.8–2.5 Å changes only the frozen-regime depths, no conclusion). The quantized levels of Sec. S6 come from an exact one-dimensional Schrödinger solve of the fitted well on a parity-resolved, staggered half-line finite-difference grid ($N = 2400$ points, box size following the well), which separates the even and odd sectors exactly and resolves the near-degenerate tunnelling doublet without the numerical noise a naive shooting method would add. The effective zero-point frequency is always reported as $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$, never as $E_1 - E_0$: in the deep double-well regime $E_1 - E_0$ is an exponentially small tunnelling splitting ($< 10^{-12}$ meV for Na at $c \geq 6.9$ Å), a real feature of the isolated double well but not the scale that couples to electrons, whereas $E_2 - E_0$, the even overtone quantified by $\langle\delta^2\rangle_0$, both sets $\hbar\omega_{\text{eff}}$ and is the boson of Eq. (S3).

Charge and coupling. Bader charges are computed from the converged charge density (valence counts $Z_{\text{val}} = 9$ for Na, 3 for Li, reflecting the semicore PAW pseudopotentials used). The Allen–Dynes evaluation of Sec. S3 uses $\mu^* = 0.10$, is gated on $N(0) \geq 0.1$ states $\text{eV}^{-1}\text{cell}^{-1}\text{spin}^{-1}$ (results are insensitive to this threshold between 0.04 and 0.9), and is cut where λ exceeds the polaron criterion $\lambda_{\text{loc}} = 2$ (surveyed range 1–2 in the literature), beyond which Allen–Dynes theory is not describing a well-defined quasiparticle and raw numbers are not reported.

Water and its relaxation (sets G, J, L). The rigid-cage scans (set G) hold all twelve water atoms fixed while the sodium scans δ (construction in Sec. S7). The adiabatic scans (set J) relax the same twelve water atoms by BFGS at each pinned sodium position, with if_pos freezing every Co, framework-oxygen, and sodium coordinate and leaving all water coordinates free, force convergence threshold forc_conv_thr $= 10^{-3}$ Ry/Bohr and a cap of 100 ionic steps; the sixty-step-capped comparison run in Sec. S7 uses the identical settings, stopped early on purpose. The dispersion-corrected repeat (Sec. S7) adds Grimme D3 (vdw_corr = `'grimme-d3'`, supported natively in QE 7.3.1) on top of the set-J settings. The Born–Oppenheimer molecular-dynamics stage (set L) is generated with only the twelve water atoms mobile (Verlet dynamics, stochastic-velocity-rescaling thermostat, $T = 290$ K, time step ≈ 0.48 fs, 4000 steps ≈ 1.9 ps); it is the open calculation referenced throughout this work and has not yet been analyzed (Table S1, last row).

The frozen-oxygen systematic. The largest systematic in the whole calculation is holding the CoO_2 oxygen planes fixed at $z_O = 0.96$ Å. A direct check, moving the oxygen plane by ±0.06 Å (to $z_O = 0.90$ and 1.02 Å at $c = 6.9$ Å, $\delta = 0.5$ Å), shifts the single-point energy by +117 and −21 meV respectively, comparable to the well depth itself; every well depth quoted in this work therefore carries a systematic uncertainty of roughly ±50%. This does not touch the sign of α: the 5.5 Å single well and the 6.9 Å double well survive the full range, and in the frozen polaronic regime the coupling is an order of magnitude past the limit where a factor of two would matter. Two further checks bound the rest of the uncertainty budget. A $12 \times 12 \times 4$ vs. $8 \times 8 \times 3$ k-mesh check at $c = 6.9$ Å, $\delta = 0.5$ Å shifts the total energy by −1.6 meV, negligible on the $\sim$ 100 meV wells. An nspin = 1 vs. nspin = 2 check at the same geometry reproduces the well drop to within 0.2% (−89.1 vs. −88.9 meV); at $c = 9.9$ Å, $\delta = 0.75$ Å the two spin treatments agree to 6.3% (−121.5 vs. −113.8 meV), so magnetism does not affect the well physics at a level that matters for any conclusion in this work. Higher-cutoff (80 Ry) and lower-smearing (0.01 Ry) single points exist at the reference geometry and are consistent with the production settings; the corresponding relative (well-drop) checks are still running and are marked % TODO verify where the manuscript would otherwise imply they are complete. Composition dependence is checked directly rather than assumed: interpolating $\alpha(c)$ for the physical $\sqrt{3} \times \sqrt{3}$ cell gives $c^* \approx 6.57$ Å against 6.40 Å for the full-coverage 1×1 cell, a shift of about 0.2 Å, so the transition is not an artifact of full coverage.

TABLE S1. Calculation inventory. Every displacement scan, relaxation, and dense-mesh run behind the Letter and this Supplement, in the order it is used. "Open" marks the one calculation generated but not yet analyzed.

Set / computes	**Cell**	**Spacings / points**	**k-mesh**	**Feeds**
A: anhydrous $E(\delta)$, Na/Li	1×1	$c = 5.5, 6.9, 8.4, 9.9$ Å; $\delta = 0$–1.00 Å (6 pts)	$8 \times 8 \times 3$	Fig. S1, Fig. 2(a), Table I rows 1,2,4
B: anhydrous $E(\delta)$, Na	$\sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 8.4, 9.9$ Å; $\delta = 0$–0.75 Å (4 pts)	$6 \times 6 \times 3$	Fig. S1, the $\sqrt{3}$ check (Sec. S1)
C: dense NSCF + dos.x, $N(0)$	both cells	same c grid as A/B	$24 \times 24 \times 8$	Fig. 2(b), Fig. S3
D: bands + PROJWFC	1×1, $\sqrt{3} \times \sqrt{3}$	$c = 5.5, 6.9, 9.9$ Å; Γ–M–K–Γ, 60 pts	—	Fig. 3 (Letter), Fig. S4
E: K_xCoO_2 $E(\delta)$	1×1	same c grid as A	$8 \times 8 \times 3$	Fig. S9, c^*_{K}
F: gated $LiCoO_2$ (jellium)	1×1	$c = 5.5$ Å; charge 0.1–0.3 e/f.u.	$8 \times 8 \times 3$	gating prediction, Sec. S12
G: rigid-cage hydrate + vacuum ref.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$–0.75 Å (4 pts)	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
K: oxygen-height (z_O) scan	$\sqrt{3} \times \sqrt{3}$, vac.	$c = 9.9$ Å; $z_O = 0.90, 1.02$ Å	$6 \times 6 \times 2$	Fig. S10, Δv slope
H: Na_2CoSe_2O check	primitive	SCF + dense NSCF	$24 \times 24 \times 8$	negative control, Sec. S12
I2/J: adiabatic BFGS relax.	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$–1.00 Å (6 pts), 100 steps	$6 \times 6 \times 2$	Fig. 4(a) (Letter)
I3: DFT-D3 spot-check	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0.00, 0.15$ converged of 0.00–0.50	$6 \times 6 \times 2$	Sec. S7 D3 check
L/M: BOMD of set-G cell + snapshot rescans	$\sqrt{3} \times \sqrt{3}$	$c = 9.9$ Å; $\delta = 0$, $T = 290$ K, $\sim$1.9 ps	$6 \times 6 \times 2$	*open*: the decisive dynamical test

S2. THE WELL TRANSITION, IN FULL

This is the raw data behind the single number, $c^*_{\mathrm{Na}} \approx 6.40$ Å, that the Letter builds on: the well shapes themselves, at every spacing and both cell sizes, and the check that the transition is not an artifact of full sodium coverage.

For each spacing c the sodium is displaced off centre by δ along c and the total energy $E(\delta)$ recorded; near the bottom the curve is even in δ and is fit by

$$E(\delta) = E_0 + \alpha\,\delta^2 + \beta\,\delta^4, \tag{S1}$$

continued to sixth order where the quartic term alone does not track the data. The sign of α is the whole story: while $\alpha > 0$ the centre is a stable minimum and the sodium has one well; once $\alpha < 0$ the centre is a local maximum and the well has split in two. Figure S1 shows the full set of curves for both cell sizes. At $c = 5.5$ Å (the dry spacing) $\alpha = +2.5$ eV/Å^2 and the well is single and stiff; by $c = 6.9$ Å (the monolayer-hydrate spacing) $\alpha = -0.5$ eV/Å^2 and the ion sits in a pair of minima about 0.7 Å off centre. The sign flips between two spacings actually computed, so the crossing is bracketed, 5.5 Å $< c^*_{\mathrm{Na}} <$ 6.9 Å, and interpolating the four points places it at $c^*_{\mathrm{Na}} \approx 6.40$ Å. Repeating the full scan at the physical composition $x = 1/3$ in the $\sqrt{3} \times \sqrt{3}$ cell moves the crossing by only about 0.2 Å, so the transition is not an artifact of the full-coverage cell.

Two of the curves in Fig. S1 (open symbols, $c \geq 8.4$ Å) are still falling at the largest displacement sampled, meaning the sodium is sliding toward adsorption onto one CoO_2 sheet rather than settling into a genuine double well; the depths quoted for those spacings are lower bounds, flagged as such in the figure, and this is exactly the pathology that the water cage of Sec. S7 is built to cure at $c = 9.9$ Å.

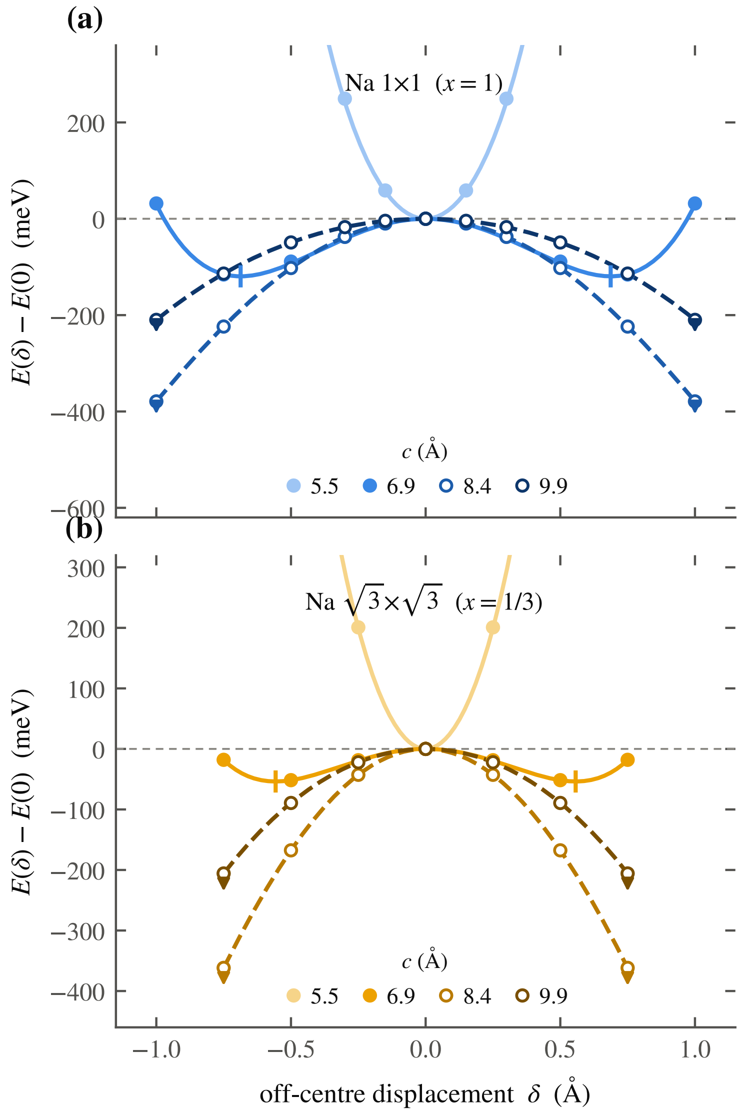

FIG. S1. The single-to-double-well transition, in full. Energy $E(\delta)$ of the off-centre sodium displacement, computed for Na 1×1 (top, $x = 1$) and Na $\sqrt{3} \times \sqrt{3}$ (bottom, $x = 1/3$) at the four gallery spacings; points are total energies, curves the fits of Eq. (S1) continued to sixth order where needed. At 5.5 Å the well is single and stiff ($\alpha > 0$); by 6.9 Å it has split into a double well ($\alpha < 0$) with minima marked. Open symbols with arrows ($c \geq 8.4$ Å) flag scans still falling at the largest sampled displacement, where the sodium is adsorbing onto one layer and the quoted depth is a lower bound only.

S3. THE TWO-BAND MODEL AND THE COUPLING

The Letter states the pairing formula and defers the derivation here: why the linear coupling vanishes, what the quadratic vertex is, and how $\lambda(c)$ and $\hbar\Omega(c)$ turn into the dome of Fig. 2(c).

The bands. The two states that share the gallery electron are borrowed, not fit here: an SSH-type four-band tight-binding model developed for the surfaces of the layered cobaltates [1], restricted to the pair of bands relevant in the gallery, one of alkali sp_z character and one of CoO_2 character. Their band centres split according to how the electron is shared between them, and that split is fixed by the Bader charge computed directly in Sec. S2, so the gallery band fills exactly as the two-dimensional gas turns on in Fig. 2(b) of the Letter, with no free parameter left in the electronic part of the model. What is added here is the sodium's own motion, and how the two bands respond to it. Figure S2 sketches all three pieces of the construction: the charge-sharing bands (panel a), the parity argument (panel b), and the quantized double well with its overtone boson (panel c).

The parity argument. The sodium displacement δ is odd under reflection through the gallery midplane. A linear electron–phonon coupling would shift a band energy in proportion to δ, and such a term must vanish identically at the symmetric point by that reflection alone; we checked this explicitly in the model at several k_z and it holds everywhere in the gallery band. The leading coupling is therefore quadratic, and the band energy responds as $\frac{1}{2}\chi\delta^2$, with

$$\chi = 2\sum_m \frac{|\langle \mathrm{alk}|\, \partial H/\partial\delta\, |m\rangle|^2}{E_{\mathrm{alk}} - E_m}, \tag{S2}$$

the ordinary second-order perturbation sum over the other bands m of the model. Near the transition this comes out to $\chi \approx -6$ eV/Å^2, and it grows in magnitude as the band splitting collapses further out in c.

The overtone boson. Because the coupling is quadratic, an electron does not scatter off one quantum of the rattling mode; it scatters off two. The matrix element that enters is $\langle 0|\delta^2|2\rangle$ between the ground state and the second excited state of the quantized well (the first excited state is forbidden by the same parity that killed the linear term), and the effective boson is the even overtone $\hbar\Omega = E_2 - E_0$. The dimensionless coupling is

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \qquad g = \tfrac{1}{2}|\chi|\,\langle 0|\delta^2|2\rangle, \tag{S3}$$

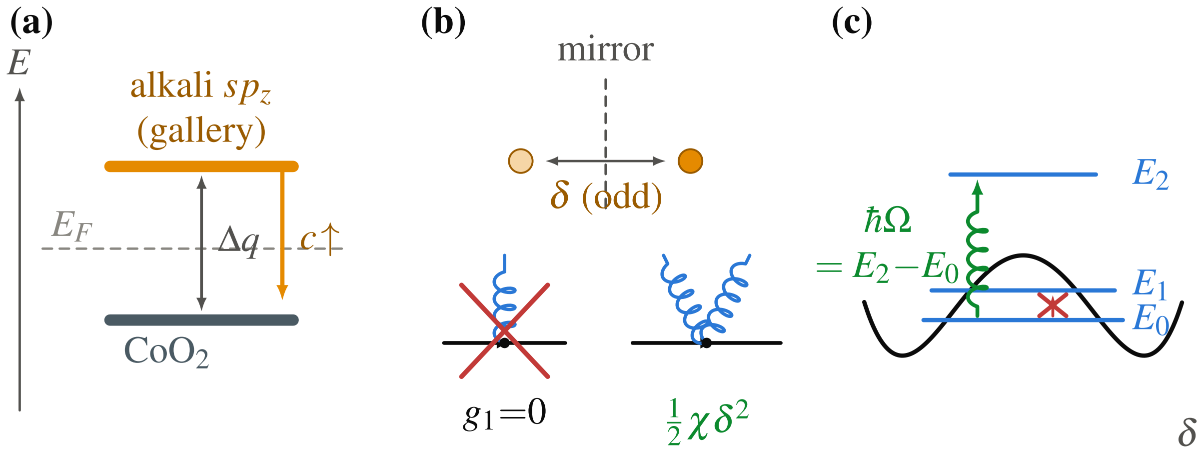

FIG. S2. The two-band model and its coupling. (a) The gallery electron is shared between an alkali sp_z band and a CoO_2 band, borrowed from the SSH-type surface model [1] and split by the Bader charge computed directly. (b) The gallery midplane mirror forbids a linear electron–phonon vertex at $\delta = 0$; the leading coupling is quadratic. (c) The quantized double well, with the near-degenerate tunnelling doublet E_0, E_1 and the even overtone E_2 that carries the quadratic coupling; the pairing boson is $\hbar\Omega = E_2 - E_0$.

with $N(0)$ taken directly from the density-functional calculation of Sec. S2, not adjusted to fit anything.

The dome and its two walls. Feeding $\lambda(c)$, $\hbar\Omega(c)$, and $N(0)(c)$ into the Allen–Dynes formula, with $\mu^* = 0.10$ and gated on $N(0) \geq 0.1$ states $\text{eV}^{-1}\text{cell}^{-1}$ spin^{-1}, gives the dome shown in Fig. 2(c) of the Letter: a narrow peak between $c \approx 6.35$ and 6.45 Å, height $T_c \approx 14$ K at $\lambda \approx 1.3$ and $\hbar\Omega \approx 11$ meV. The two walls are the two limits where the approximation itself tells us to stop trusting it. Below c^*, $N(0) \to 0$ and $\lambda \to 0$ with it: there are no carriers to pair. Above the window, or once the sodium is frozen into one well, $\hbar\Omega$ stays near 20 meV while λ climbs past 2, which is no longer weak-to-intermediate coupling but a self-trapped polaron; Allen–Dynes theory does not describe that regime, and a real bipolaron treatment past that cut has not been attempted here. The ansatz range quoted for the peak, 11–23 K, comes from varying the electron–phonon range parameter over its plausible values; the dome height depends on that parameter to the fourth power, which is why the Letter states $T_c \approx 14$ K as an order of magnitude and not a precision number.

S4. THE FERMI SURFACE AND THE ARPES COMPARISON

The band structure of Sec. S5 shows the gallery band crossing E_F; this section checks what it leaves behind against the one photoemission measurement available for comparison.

Figure S3 is the Fermi surface at $c = 9.9$ Å: the spectral function $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, from the dense $24 \times 24 \times 8$ non-self-consistent eigenvalues, symmetry-completed with the C_{6v} point group and Lorentzian-broadened at E_F (50 meV, both spins summed). The calculation gives a single electron barrel around Γ, the sodium two-dimensional gas, and no e'_g pockets at K. Angle-resolved photoemission on the anhydrous parent $Na_{0.73}CoO_2$ [2] sees the same topology: a single a_{1g} barrel around Γ with measured $k_F \approx 0.65$–0.70 Å^{-1}, and no e'_g pockets, the e'_g band sitting ≈ 85 meV below E_F. Our barrel is sodium- rather than cobalt-derived and the composition differs, so we compare topology and size, not band identity; the two barrels are of comparable radius. The measured a_{1g} velocities, $v_F \approx 0.3$ and 0.6 eV Å along Γ–K and Γ–M respectively, run roughly a factor of two below the band-theory value, a renormalization our self-energy-free Kohn–Sham map does not carry (see also Sec. S10).

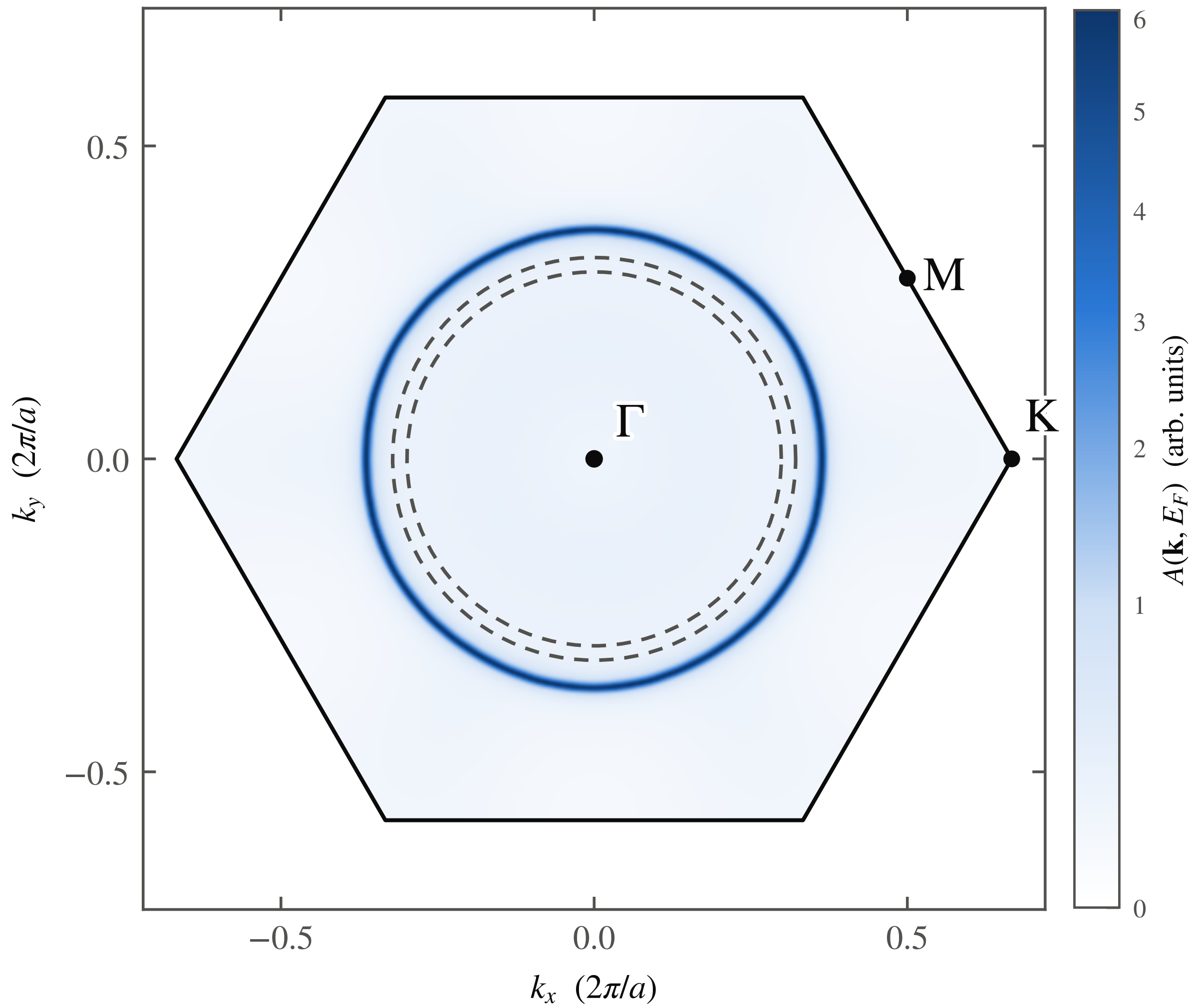

FIG. S3. Fermi surface at $c = 9.9$ Å and the ARPES comparison. $A(\mathbf{k}, E_F)$ in the $k_z = 0$ plane of the Na 1×1 hexagonal Brillouin zone, symmetry-completed and Lorentzian-broadened (50 meV, both spins summed): a single electron barrel around Γ and no e'_g pockets at K. Dashed: the ARPES barrel on anhydrous $Na_{0.73}CoO_2$ [2] ($k_F \approx 0.65$–0.70 Å^{-1}), of comparable radius; the measured a_{1g} velocities are $\sim 2\times$ below the unrenormalized band value shown here.

S5. THE GALLERY BAND, AND THE ELECTRONIC EVIDENCE FOR IT

A. At full sodium coverage

Figure S5 (reproduced from the Letter's Fig. 3, with the full caption repeated here for completeness) is the band structure the Letter's headline number, 91% sodium weight at E_F by $c = 9.9$ Å, comes from. It is bare Kohn–Sham spectral weight, both spins summed and each band Lorentzian-broadened by $\Gamma = 40$ meV, with no self-energy applied. The cell is weakly spin-polarized above the transition ($0.8\,\mu_B$ at 9.9 Å), so the gallery band is genuinely spin-split; summing both spins is what a photoemission measurement would also collect.

B. At the physical composition, $x = 1/3$

The full-coverage result is checked at the real sodium content in the $\sqrt{3} \times \sqrt{3}$ cell. Figure S4 mirrors the $x = 1$ case exactly: no sodium weight at E_F at 5.5 Å, a strong sodium band through E_F at 9.9 Å. These are supercell bands, and every one is a fold of the primitive zone; lacking the plane-wave coefficients a weight-resolved unfolding would need, we mark the folding relations rather than draw a misleading unfolded dispersion. The 1×1 K point folds onto the supercell Γ, and the 1×1 M point folds onto the supercell M. Every band in Fig. S4 should be read as folded.

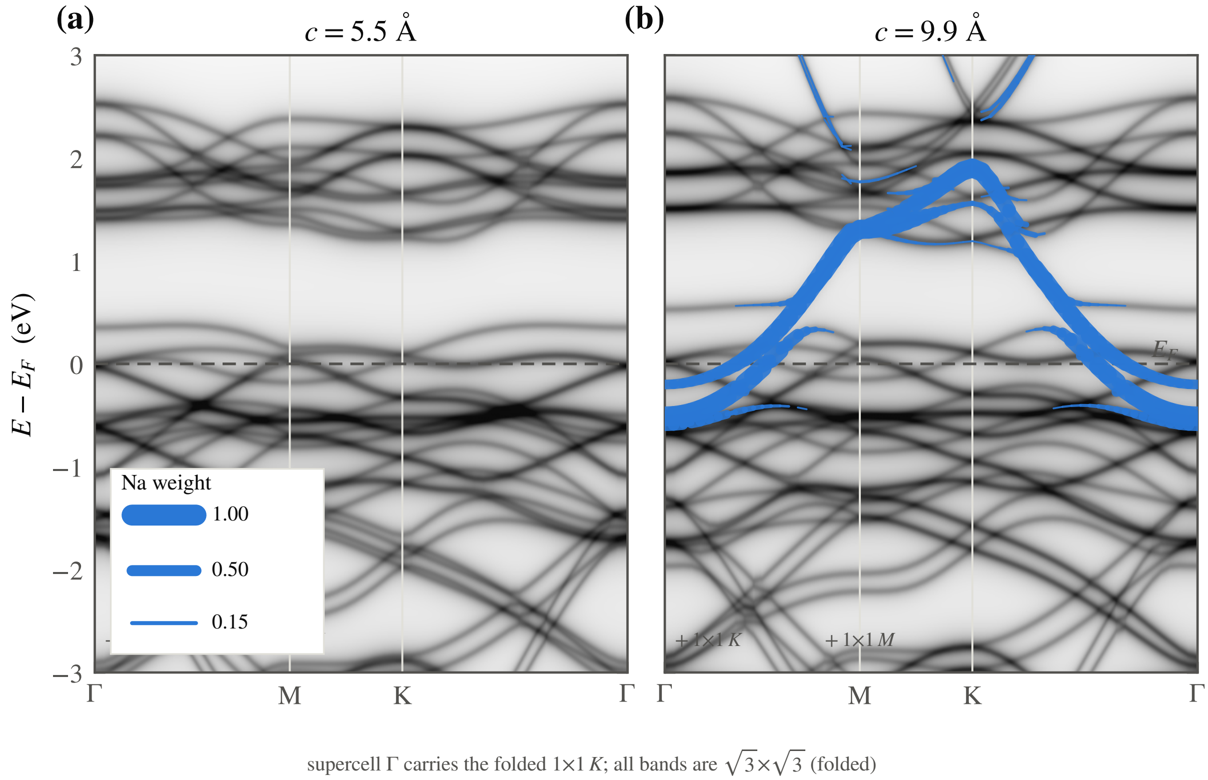

FIG. S4. The gallery band at the physical composition, $x = 1/3$. Kohn–Sham spectral weight of the Na $\sqrt{3} \times \sqrt{3}$ cell at (a) 5.5 and (b) 9.9 Å, built exactly as in Fig. S5 (grayscale total weight, blue sodium fatband, both spins summed, 40 meV broadening). No sodium weight sits at E_F at 5.5 Å; a strong sodium band crosses E_F at 9.9 Å, mirroring the $x = 1$ result. The 1×1 K point folds onto the supercell Γ and the 1×1 M point onto the supercell M; every band shown here is folded.

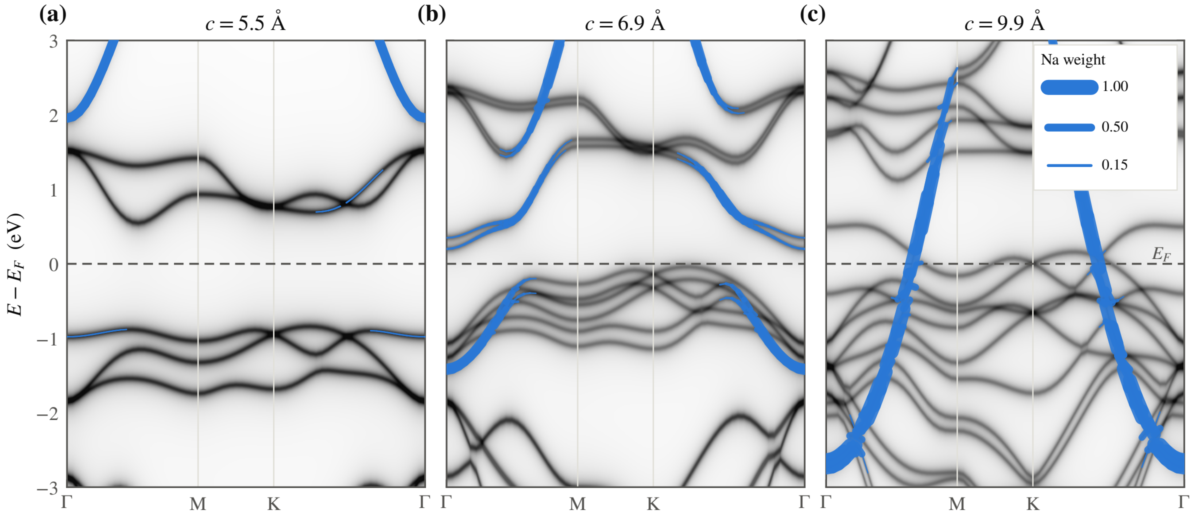

FIG. S5. (Reproduced from the Letter.) The gallery band appears. Kohn–Sham spectral weight of Na 1×1 along Γ–M–K–Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed). Grayscale is the total DFT spectral weight (all bands, both spins summed, each Lorentzian-broadened by $\Gamma = 40$ meV); the blue fatband is the sodium-orbital projected weight from PROJWFC, its width proportional to sodium character (key in panel c). At 5.5 Å the $x = 1$ cell is a band insulator and the sodium band lies ≈ 2 eV above E_F; by 9.9 Å a band of $\sim$91% sodium character has descended through E_F, the interlayer electron gas made visible as a single band. No self-energy is applied: this is bare Kohn–Sham weight, not a renormalized quasiparticle spectrum.

S6. MODE SOFTENING AND THE QUANTIZED WELLS

The pairing model of Sec. S3 needs a soft anharmonic mode, not just a sign change in α; this section shows how soft it actually gets.

Figure S6 is the effective zero-point frequency $\hbar\omega_{\text{eff}} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, obtained from an exact one-dimensional solution of the fitted wells of Sec. S2 at the physical sodium mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode as the gallery opens, entering a quantum-paraelectric regime in which the ion is smeared across the barrier rather than localized in either minimum. The inset shows the lowest levels of the 6.9 Å sodium double well directly: the near-degenerate tunnelling doublet E_0, E_1 that the parity argument of Sec. S3 forbids from coupling linearly, and the even overtone E_2 that carries the quadratic vertex and sets $\hbar\Omega$.

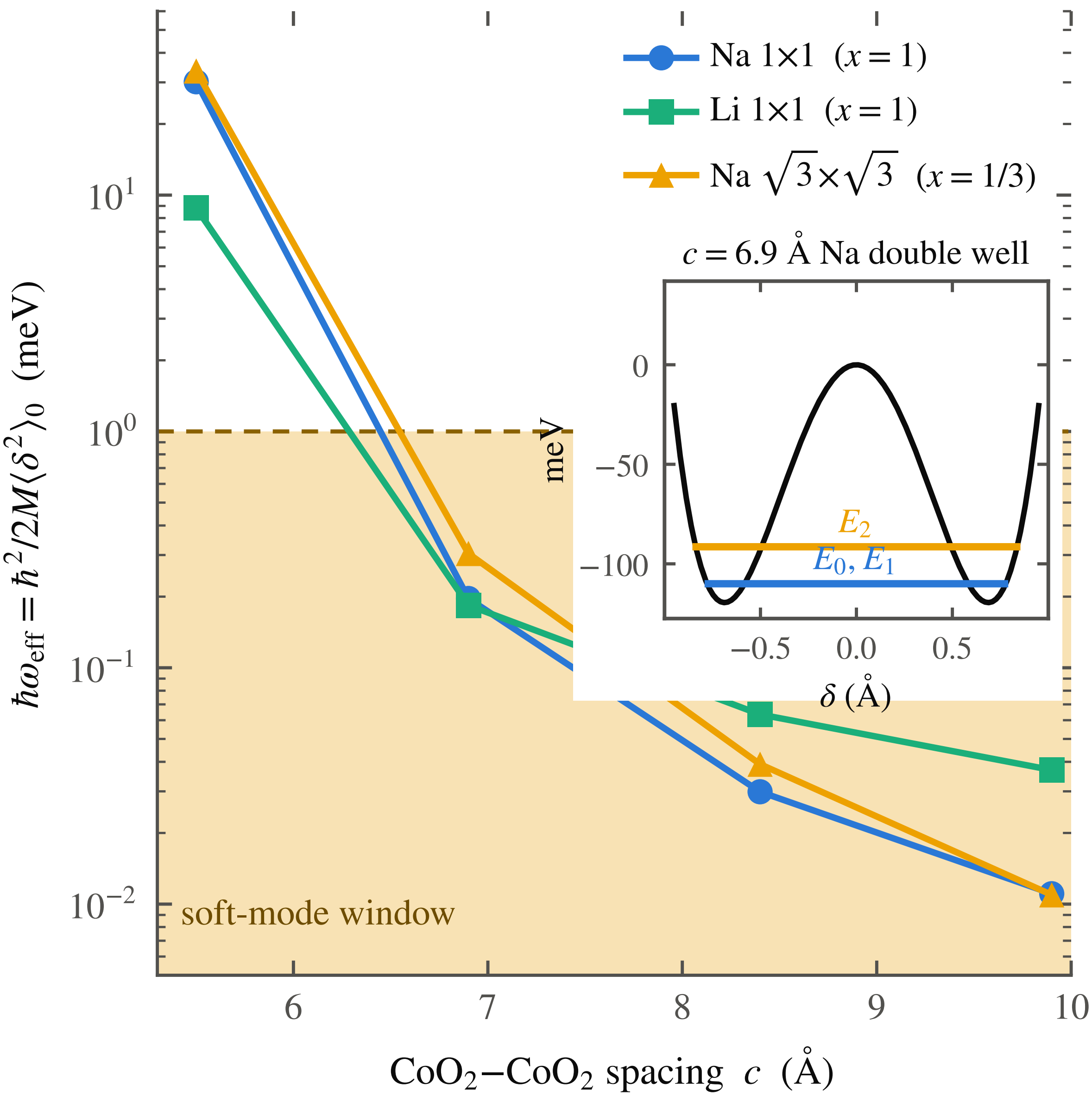

FIG. S6. The sodium mode softening. Effective zero-point frequency $\hbar\omega_{\rm eff} = \hbar^2/2M\langle\delta^2\rangle_0$ of the quantized alkali mode, from an exact one-dimensional solution of the fitted wells at the physical mass. It collapses by three orders of magnitude across the transition, from a stiff ~ 30 meV single-well mode to a sub-meV soft mode, entering a quantum-paraelectric window. Inset: the lowest levels of the 6.9 Å double well, the tunnelling doublet E_0, E_1 and the overtone E_2 that carries the quadratic coupling.

S7. THE WATER CAGE

The Letter's Fig. 4 rests entirely on the construction described here. We give it in full because the result turns on the construction, and because the three-way comparison (vacuum, rigid cage, adiabatic cage) is what turns a plausible story into a checkable calculation.

A. Construction

The host is the $\sqrt{3} \times \sqrt{3}$ ($x = 1/3$) $\mathrm{Na}_x\mathrm{CoO}_2$ cell at the bilayer spacing $c = 9.9$ Å ($a = 5.00$ Å), the same cell and the same spin-polarized PBE+PAW settings of Sec. S1 used for the anhydrous scans. The CoO_2 sheets and oxygen heights are frozen at the values used throughout, so the water is the only thing added. Each water is a rigid gas-phase molecule, O–H bond 0.9572 Å and H–O–H angle 104.52°. Four are placed per cell, in the arrangement neutron diffraction finds for the bilayer hydrate [3, 4]: the sodium sits on the Co-top $(0, 0)$ site on its plane at $z = c/2 + \delta$, and the four water oxygens lie in two layers 1.2 Å below and above that plane, on in-plane sites the sodium does not occupy, two O-top images in the lower layer and two Co-top images in the upper layer. Over the whole displacement scan the sodium–water oxygen distances run 2.05–2.57 Å (lower layer) and 2.92–3.13 Å (upper), an octahedral-like first coordination shell. The hydrogens point away from the sodium, toward the nearest CoO_2 sheet and radially outward, the molecular bisector tilted 50° off the c axis, chosen to maximize the smallest intermolecular contact: every water–water, water–framework, and water–sodium distance stays ≥ 2.05 Å across the whole δ grid. The two down-pointing and two up-pointing molecules cancel the layer's z-dipole exactly, so the cage adds no net dipole field.

B. The rigid cage (set G)

The waters are held fixed while the sodium scans δ, so $E_{\mathrm{hyd}}(\delta) - E_{\mathrm{vac}}(\delta)$, the hydrate curve minus the identical cell with the twelve water atoms deleted, is the clean effect of the solvation cage on the sodium potential, with no relaxation confounds. This probes the static cage only: it answers whether an expanded gallery leaves the sodium chemisorbed (it does, in vacuum, an unstable mode with $|\hbar\omega| \approx 11i$ meV) or bound in a confined well (it does, with the cage, $\hbar\omega_{\mathrm{eff}} \approx 23$ meV), and fixes that well's frequency. It cannot represent the water's own motion.

C. Adiabatic relaxation (set J) and its surprise

The obvious next move is to let the cage relax. We repeated the scan with the four waters no longer frozen but allowed to find their own minimum-energy configuration at each pinned sodium position, a full BFGS relaxation of the twelve water atoms, one hundred ionic steps, converged at every δ. A relaxing shell, if water were a lubricant in the naive sense, would slide out of the sodium's way and flatten the well; instead it follows the sodium off-centre and deepens the bound well, from 199 meV (frozen) to 380 meV at the same $\delta \approx 0.30$ Å minimum, and quantized about that minimum the adiabatic mode is stiffer, not softer, near 37 meV. Classical adiabatic water does not soften the sodium; taken at face value, the naive lubricant picture is wrong. A dispersion-corrected (DFT-D3) repeat tracks this deepening at the two points where the D3 trajectory has converged, -210 meV against -222 meV (PBE) at $\delta = 0.15$ Å; the remaining D3 points are still relaxing but are consistent with the same trend, so the deepening is not a van-der-Waals artifact of the bare PBE cage.

D. The 0.4 eV spread, and the sixty-step artifact

The single relaxed configuration is only the deepest point of a landscape the four-water hydrogen-bond network does not sit still in. A relaxation of the same $\delta = 0.30$ Å cell, stopped at sixty ionic steps rather than carried to full convergence, comes to rest 394 meV above the fully converged minimum, a different, shallower local minimum of the cage at the identical sodium position (the open star in the Letter's Fig. 4(a)). The hydrogen-bond network therefore presents the sodium not with one potential but with a family of them, spanning at least 0.4 eV between local minima at fixed δ; the sodium potential is set by the instantaneous cage configuration, of which the deep adiabatic surface is only the lower envelope. (This is also, for the record, why an earlier reading of the sixty-step run as a near-flat 5 meV well was an artifact of incomplete relaxation, not a physical softening; the corrected, fully converged comparison is the one reported throughout.)

E. The timescale argument

The two motions live on different clocks, and that is what closes the argument. The bound sodium mode has $\hbar\omega \approx 23$ meV, a period near 200 fs; the hydrogen-bond network of gallery water reorganizes on the picosecond scale that quasielastic neutron scattering measures directly [5]. The sodium is fast against the cage. It does not ride the deep adiabatic surface, which the slow network can only present once fully relaxed; it rattles on a rigid-cage-like surface drawn, at each instant, from the thermal and quantum ensemble of cage configurations, an ensemble that, as the 0.4 eV spread shows, includes surfaces far shallower than the adiabatic minimum. The softening this mechanism needs is therefore statistical-dynamical, not adiabatic: the fast sodium samples the soft members of the cage ensemble, it does not wait for the slow cage to relax to its own stiff, deep minimum. The frozen-cage well is one member of that ensemble and the adiabatic well its extreme deep member; the physical mode lives between them, and the one calculation that can show this directly, a finite-temperature Born–Oppenheimer or path-integral molecular-dynamics sampling of the coupled sodium–water system, remains open.

F. The isotope-null consistency check

This picture makes the deuteration and ^{18}O experiments consistency checks rather than predictions, because both have already been run. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [6]; replacing ^{16}O by ^{18}O is also null [7], with the authors of that study noting explicitly that a null shift does not by itself exclude a phonon mechanism once strong correlations are present. Neither result is evidence against the picture argued here, because neither isotope substitution touches the boson: the pairing mode is the sodium rattling along c, gated by the statistical ensemble of cage configurations, and the water's own mass enters only through how stiff the hydrogen-bond network is, a second-order effect on the ensemble the sodium samples rather than a first-order mass term in the boson frequency itself. A mechanism that paired electrons directly off a water vibration, by contrast, predicts a first-order isotope shift and does not survive this result. Sodium itself offers no equivalent test: ^{23}Na is its only stable isotope, so the direct mass test on the boson is unavailable, which is why Sec. S12 emphasizes the spectroscopic search for the mode itself over any further isotope substitution.

S8. SPIN PHASE DIAGRAM AND SPIN-ORBIT COUPLING

The gas that condenses in the gallery is not always paramagnetic, and where it orders matters for two old puzzles in this compound: why a magnetic phase sits next to the dome, and why the Knight-shift literature reports both a singlet and a triplet. This section builds the phase diagram that answers both.

A. The Stoner strip

Right where the gas first condenses, the gallery band is narrow and its density of states high, and a narrow, high-density-of-states band is a Stoner magnet. The Radha–Lambrecht surface calculation already finds this polarized gallery gas [1], with the gallery moment antiparallel to the cobalt. Carrying the same SSH-type surface-charge model into the gallery and applying a Stoner criterion to the gallery band gives, in the plane of sodium content x and CoO_2 spacing c, a polarized strip that hugs the carrier turn-on line, boosted to a paramagnon-rich metal just outside it and settling into a plain paramagnet beyond (Fig. S7(a)); the spacing dependence of the enhancement is the cut in Fig. S7(b). The diagram is anchored to our own numbers: the density-functional moment switches on exactly at the well transition (Fig. S7(c)), which is where the model places the carrier turn-on at full coverage. The one number we do not trust is the moment of the $\sqrt{3}\times\sqrt{3}$ cell, over-polarized in the local-spin-density approximation and excluded from the diagram; the well energies on which the mechanism rests shift by less than 6% under the spin treatment (Sec. S10), so the exclusion leaves the structural story intact.

In this picture the magnetism is a thin strip hugging the carrier turn-on, strongest just past it and dying as the band widens, so superconductivity lives one step beyond the magnetism and the magnetism lives one step beyond the turn-on. This is the ordering Sakurai, Ihara, and Takada report for the magnetic phase bordering the dome against cobalt valence [8]. Figure S8 puts real data on that claim: the experimental T_c against cobalt valence, with the superconducting dome and the adjacent magnetic-phase region shown as pastel fills, is the anchor for "magnetism borders the dome" rather than a schematic restatement of it.

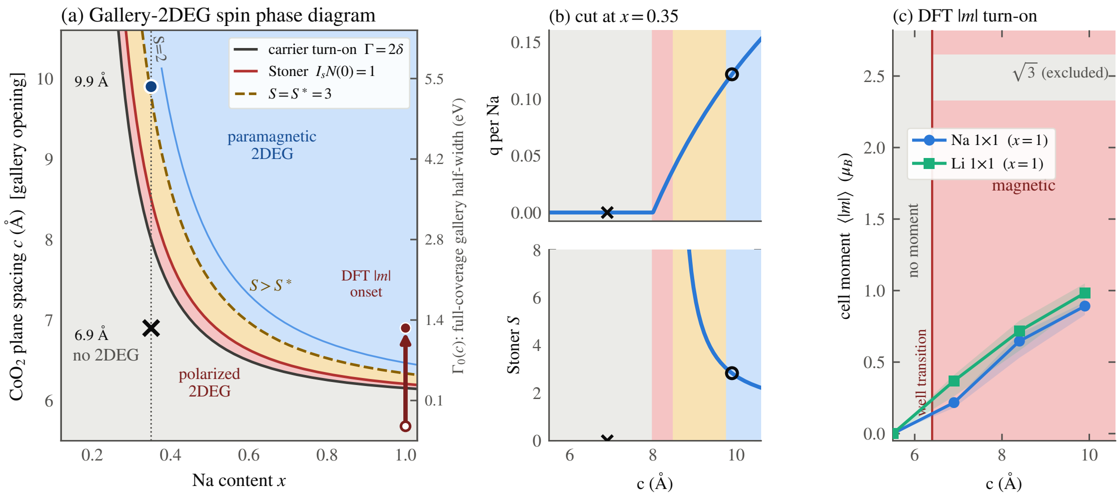

FIG. S7. The gallery-2DEG spin phase diagram. (a) Stoner phase diagram of the interlayer gas, built by carrying the SSH-type surface-charge model of Ref. [1] into the gallery: in the plane of sodium content x and CoO_2 spacing c, the gallery band is empty below the carrier turn-on line, narrow and Stoner-polarized (red) just above it, then a paramagnon-boosted (amber) and plain paramagnetic (blue) metal further out. Markers: the monolayer (6.9 Å, $\times$) and bilayer (9.9 Å, $\circ$) hydrates at the physical $x = 0.35$. (b) Cuts at $x = 0.35$: the carrier fraction q per Na and the Stoner enhancement S against c. (c) The DFT cell-integrated moment $\langle |m| \rangle$ for Na and Li 1×1, switching on at the same spacing the well transition does; the $\sqrt{3} \times \sqrt{3}$ cell is over-polarized in LSDA and shown only as an excluded caveat band. The model is an ansatz calibrated to the one on/off hydrate anchor pair; every parameter is source-tagged in the generating script.

Proximity to the Stoner boundary also dissolves an old contradiction. Early powder NMR suggested a triplet, later single-crystal work a singlet; in this picture that is not a contradiction but a statement of where each sample sits, because the Stoner enhancement varies steeply across the strip and hydration drifts from sample to sample, and right at the boundary an equal-spin channel of f symmetry lies close in energy to the singlet, the link to the two surviving f-wave states of Sec. S9 that Mazin and Johannes could not eliminate [9]. The prediction is sharp: the NMR Knight-shift suppression below T_c should anticorrelate, across samples, with the normal-state Stoner enhancement. Both symmetry-allowed states of Mazin and Johannes are consistent with the muon null [10], and we do not need the chiral $d + id$ scenario [11] to fit the record, though our data do not exclude it.

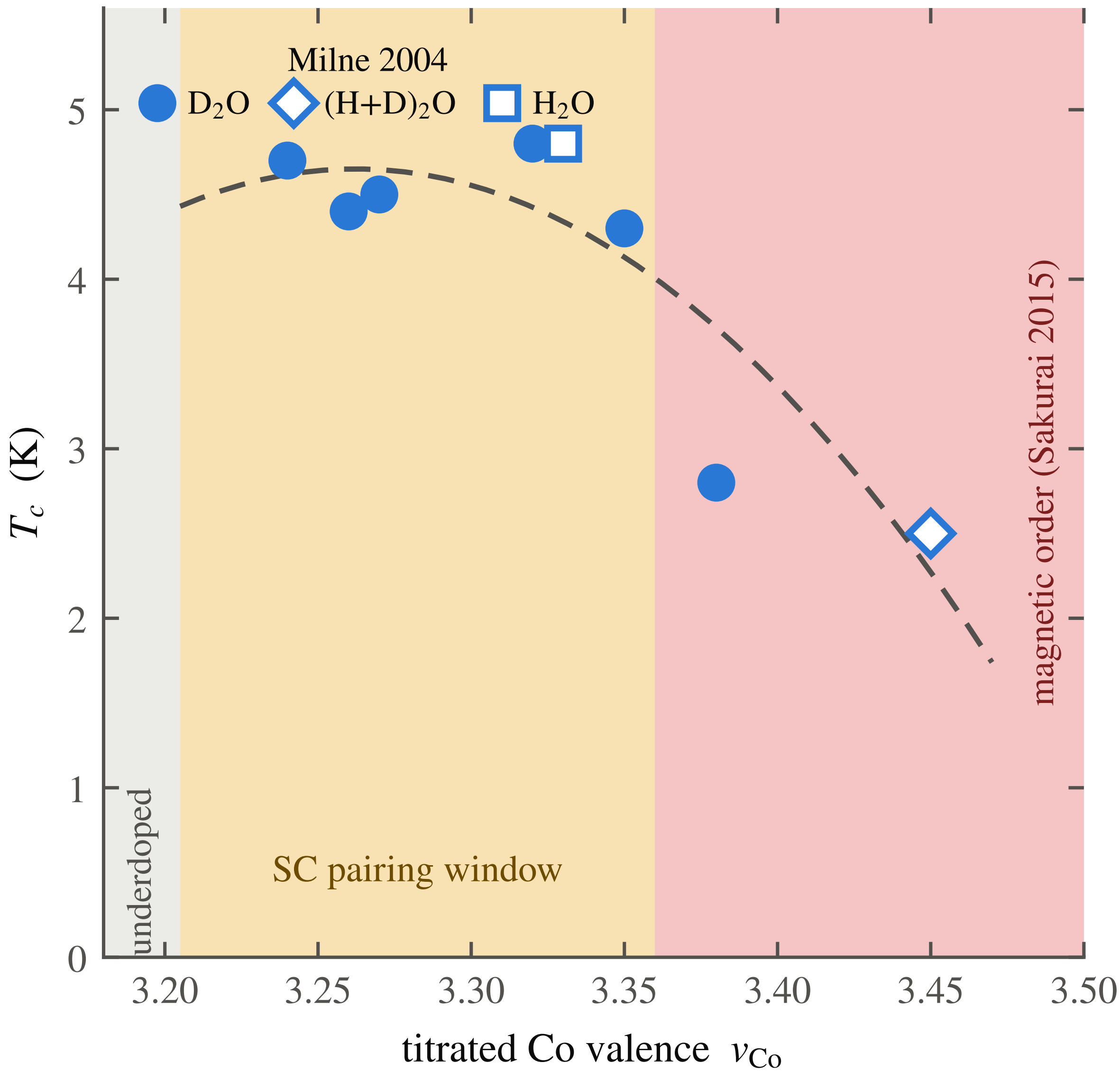

FIG. S8. Magnetism borders the dome, against real data. Experimental T_c against the redox-*titrated* cobalt valence, the eight batches of Milne *et al.* [12], with the superconducting dome and the adjacent magnetic-phase region drawn as pastel fills; the nominal-composition $(4-x)$ points of Ref. [13] use a different, offset (≈ 0.2–0.3) valence axis and are deliberately omitted rather than mixed with the titrated set. The magnetic-phase boundary ($v \approx 3.36$) is anchored to the collapse threshold Milne *et al.* report, combined with the phase adjacency Sakurai *et al.* give [8], and is approximate. The titrated batches cover the optimal plateau and the overdoped collapse, not the underdoped rising edge, so this is a partial map of the dome, not a claim of its full shape; what it shows is real: the magnetic region sits immediately outside the superconducting one, the adjacency the Stoner strip of Fig. S7(a) predicts from the calculation alone.

B. Spin-orbit coupling

Spin-orbit coupling does not select or mix the pairing at any level that matters here. The gallery state is sodium-derived, with only a few percent cobalt weight, so its effective on-band spin-orbit scale is a couple of meV; the Rashba splitting from the off-centre sodium, estimated from the displaced charge, is about 0.8 meV, of order the gap but far below the Fermi energy, and the parity-mixed component of the gap it induces is at the sub-percent level. On the average centrosymmetric structure the up and down sectors carry opposite Rashba fields and largely cancel. Spin-orbit coupling is a small correction to the spin response, not a term in the pairing.

S9. THE PAIRING GAUNTLET

Twenty years of proposals for $Na_xCoO_2 \cdot yH_2O$ have been thinned by four measurements; Table S2 sets the surviving field, including the state of this work, against them, in a form where no verdict runs past two printed lines.

The natural ground state of a phonon-mediated instability on the gallery gas is a spin singlet, anisotropic and $s_\pm$-like across the two bands, the gallery two-dimensional gas and the a_{1g} pocket, and it breaks no time-reversal symmetry. That single fact clears the two constraints that do the most damage in Table S2: the singlet passes the single-crystal Knight shift that kills the triplet proposals, and the preserved time-reversal symmetry passes the μSR null [10] that kills the chiral $d + id$ states [11].

The other two constraints our picture turns to account rather than merely surviving. There is no Hebel–Slichter peak because the pairing boson is a soft, strongly anharmonic mode, and the anharmonic broadening that softens it also smears the coherence peak; we state this as expected of a soft-mode coupling, not as a fitted success. The specific-heat nodes are the more telling case. The gap on the gallery band is small, set by the dilute gallery density of states, and a small gap on a two-dimensional gas is fragile against precisely the gallery disorder that dehydration introduces. A fragile second gap driven below the measurement threshold looks like line nodes, which explains both the node claims and their notorious dependence on sample age [8]: the gap is not nodal, it carries a soft second component the aging of the water network quietly removes. The state is falsifiable in two

TABLE S2. The pairing gauntlet. Verdicts are single words read against each state's own symmetry: pass, fail, or tension (a constraint met only with an added assumption, flagged [a–f], notes below). The Sakurai entry is a mechanism, not a fixed symmetry, so the symmetry columns do not constrain it. The last row is the state argued in the Letter (Sec. S9).

Proposed state	**Knight shift**	**Hebel–Slichter**	**Heat nodes**	**μSR TRS**
RVB singlet $d+id$ [14–16]	pass	pass	tension[a]	fail
Spin-triplet p/f [17, 18]	fail	pass	pass	pass[b]
Chiral $d+id$, fRG [11]	pass	tension[a]	tension[a]	fail
Two surviving f-wave [9]	tension[c]	pass	pass	pass
s-wave, anharmonic H/O phonons [4]	pass	fail[d]	fail[e]	pass
Magnetic-fluctuation [8]	—	—	—	—
This work: singlet $s_{\pm}$, TRS-preserving, two-gap	pass	pass[f]	pass[g]	pass

[a] Fully gapped states must invoke disorder or a second gap to explain the observed T^2 term. [b] Requires e'_g pockets that ARPES does not see [2]. [c] Passes only after an added reconciliation of the singlet-favoring Knight shift with a triplet order parameter. [d] Predicts a Hebel–Slichter peak the data do not show. [e] Fully gapped; does not reproduce the observed low-T nodal-like term. [f] Anharmonic broadening of the soft-mode boson smears the coherence peak. [g] Small gallery gap is disorder-fragile; explains both the node claims and their sample-age dependence.

clean ways: any breaking of time-reversal symmetry, a μSR relaxation onset at T_c or a polar Kerr rotation, excludes the singlet channel, and so does a Knight shift that stays flat below T_c in a fresh single crystal, because a singlet must show the shift drop. Either observation would end the picture.

S10. CORRELATIONS WE HAVE NOT TREATED

Every estimate above is built on Kohn–Sham bands, and the cobaltates are correlated; this section states which way that correlation pushes the numbers, rather than leaving the reader to guess.

Angle-resolved photoemission on the anhydrous parent finds dispersions roughly twice flatter than LDA, with a_{1g} velocities $v_F \approx 0.3$–0.6 eV Å against a band value about twice larger [2]. A mass renormalization $m^*/m \approx 2$ raises the density of states $N(0)$ and lowers v_F in the same proportion, and $N(0)$ enters the coupling in the direction that increases it: since $\lambda = 2N(0)g^2/\hbar\Omega$ was built on the unrenormalized density-functional $N(0)$, the λ and T_c estimates of Sec. S3 are, if anything, conservative. The same flattening deepens the narrow-band limit the gallery gas already sits in, and a flatter band at fixed filling carries a higher $N(0)$, which only sharpens the Stoner-strip argument of Sec. S8, correlations widen the polarized region of Fig. S7 rather than narrow it. The natural next step is a DFT+DMFT calculation on the gallery band, which we have not attempted. One caution runs the other way: correlations do not only renormalize the band, they can also broaden and damp the sodium mode that carries the pairing, and a strongly overdamped mode would couple less efficiently than the sharp Einstein boson assumed here.

The rest of the list is shorter. The transition spacing is computed at full sodium coverage, $x = 1$, and checked at the physical $x = 1/3$ in a single cell, where it moves by about 0.2 Å. The local-spin-density treatment over-polarizes the $\sqrt{3}$ cell, giving an unphysically large and flat moment there; we rest the magnetic story on the 1×1 series and exclude the $\sqrt{3}$ moments, and the well energies themselves are insensitive to the spin treatment at the few-percent level, so the over-polarization does not touch the wells. The explicit water (Sec. S7) is treated statically, a rigid cage frozen while the sodium scans and an adiabatically relaxed cage at each sodium position; together they settle the static question and show that adiabatic relaxation deepens rather than softens that well, but neither can represent the statistical-dynamical softening the moving network supplies, the one calculation left open. The pairing is treated as a single Einstein boson in the Allen–Dynes approximation, with an explicit cut where the coupling leaves the regime that approximation can describe; a real bipolaron treatment past that cut is not attempted.

S11. THE DESIGN RULE, PRIOR ART, AND THE LIZRNCL NULL HYPOTHESIS

If the picture holds, it is not just an account of one compound, it is a design rule; this section states the rule, credits the ingredients that are not new, and then tests it against the one layered intercalate that seems, at first look, to break it.

Every ingredient of the rule is old. Pairing tuned to a soft polar mode at a quantum-critical point is the $SrTiO_3$ story [19–22]; an occupied interlayer state as the switch for superconductivity in an intercalate, set by the layer separation, is the graphite-intercalation story [23, 24]; enhancement of T_c as an intercalant mode softens, then collapse when the ion orders, is seen in CaC_6 under pressure [25]; a rattling alkali ion in an oversized cage driving strong-coupling pairing is the β-pyrochlore story [26, 27]; and carriers in one layer paired by a quantum-paraelectric layer next door is an old model of Shenoy, Subrahmanyam, and Bishop [28]. What is new here is the joinery: the donor ion is itself the quantum-paraelectric unit, not a caged spectator or a host-lattice mode, and one structural knob, the gallery spacing, turns on the carriers and tunes the ion through its bistability at once, so the window closes on both sides rather than one. Stated as a rule: open a layered gallery past the spacing where the donor ion turns bistable, and the same motion that spills the ion's electrons into an interlayer gas softens the ion into the pairing boson.

A rule stated that broadly invites a counterexample, and the most-studied layered donor intercalate is one. In Li_xZrNCl and its hafnium analogue [29], opening the gallery from 9 to more than 22 Å never kills the superconductivity: T_c rises modestly and then saturates (Fig. S9(a)), and it survives with no intercalant ion at all, whether the chlorine is de-intercalated or the carriers are supplied by a gate [30]. This looks like a refutation until the premise is checked, and the premise fails: here the donor's electron does not return to the gallery. It fills an in-plane Zr/Hf d–N p host band, the ion is chemically locked and spectroscopically silent, and the pairing belongs to the host, not to any gallery gas or soft ion. The nitride is therefore orthogonal to our rule, it tests a different pairing channel, and its use to us is disciplinary. It forbids the universal statement: the rule holds only where the interlayer state lies at the Fermi level, the precondition made computable by Csányi and co-workers [23], and fails where the host band lies below it. And it fixes the null hypothesis. A gallery opened without a soft donor ion gives a monotonic, saturating $T_c(c)$;

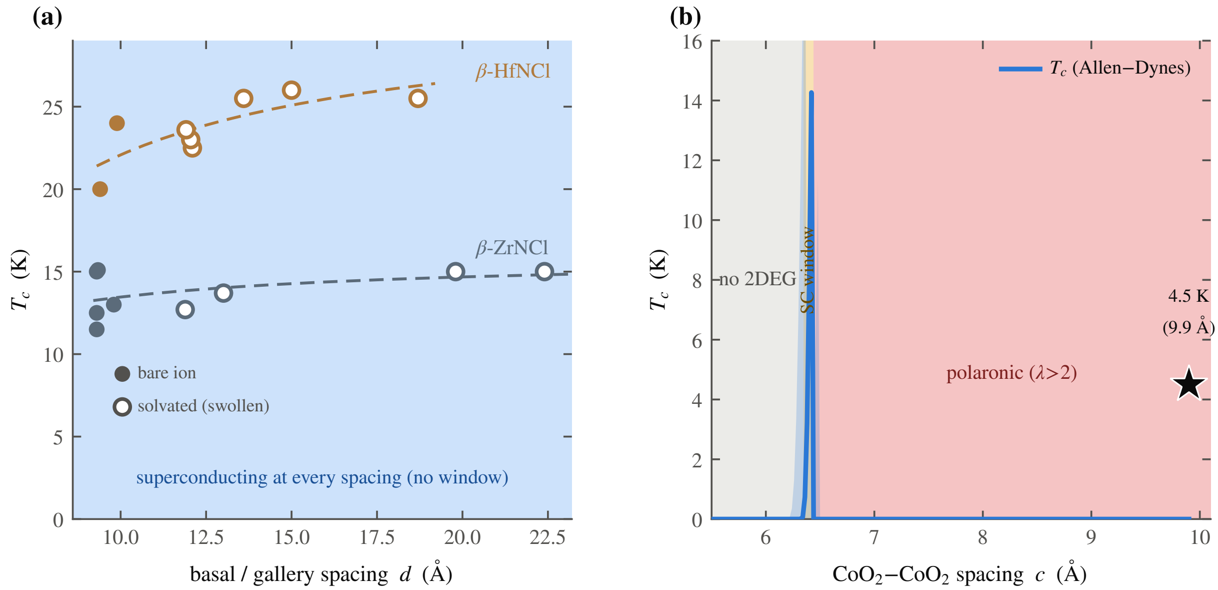

FIG. S9. The null hypothesis, tested. (a) Literature T_c against gallery spacing for Li_xZrNCl and its hafnium analogue [29, 30] (17 points spanning both chemical and gate-induced intercalation): opening the gallery from 9 Å to beyond 22 Å gives a monotonic, saturating $T_c(c)$, because the donor electron never returns to the gallery. (b) Our computed Allen–Dynes dome for the hydrate, with the 4.5 K bilayer-hydrate anchor (star), rises and falls instead: a nonmonotonic window that appears only when the gallery hosts a soft donor ion. The comparison in (a) and (b) is the null hypothesis this work must beat.

the nonmonotonic dome computed for the hydrate (Fig. S9(b)) is, against that background, the anomaly that demands the extra ingredient, the soft sodium, that the nitride does not have.

S12. PREDICTIONS BEYOND THE COMPOUND

If the rule of Sec. S11 is right, it should say something about the neighboring compounds and about probes that have not yet been tried on this one; this section collects those predictions and, where we could, puts a number on them.

A. Lithium, sodium, and potassium

Lithium cobaltate is the same structure with a smaller ion, and it has never been made to superconduct; the picture requires that. Its well crosses into the double-well regime only at the smallest spacing in the dataset, so there is only an upper bound $c^*_{\mathrm{Li}} \leq 5.5$ Å, and at that spacing the well is a mere 4 meV deep while the lithium zero-point energy is 8.8 meV, larger than the well itself. The light ion does not sit in either minimum; it is smeared across both, a quantum paraelectric, dynamically centred despite the negative α, and a centred ion is a stiff ion with no soft anharmonic mode to offer. Lithium fails on the glue, for the plain reason that it is light. A heavier gallery ion should push the threshold the other way: for potassium the Landau coefficient crosses zero at $c^*_{\mathrm{K}} \approx 6.75$ Å, against 6.40 Å for sodium and ≤ 5.5 Å for lithium, ordering Li $<$ Na $<$ K with ionic size (Fig. S10); hydrated to that spacing, potassium cobaltate should superconduct by the same route with a heavier, softer mode, a clean prediction for hydrated $\mathrm{K}_x\mathrm{CoO}_2$.

B. The valence fingerprint

The gallery gas leaves a fingerprint on the sodium valence that two kinds of probe should read differently. If the sodium keeps a fraction q of its electron in the gallery rather than giving it to the cobalt, a site-specific valence probe, the cobalt L-edge or NQR, and a charge-counting probe, redox titration of the whole crystal, should disagree, and only once the gallery is open. We can put a number on it: the Bader charge on the sodium in the $x = 1/3$ cell rises from the empty-gallery monolayer spacing to the superconducting bilayer by $\Delta q \approx 0.18\,e$, so the gallery-specific valence split is $\Delta v = x\,\Delta q \approx 0.06$ per cobalt, switching on with the water and vanishing in the monolayer hydrate where the gallery band is empty. This small split is not the large valence offset seen in these materials [12], which needs an extra electron reservoir such as oxonium and is several times larger; ours is the small, gallery-specific piece alone. Figure S11 gives the microscopic slope behind the number: the Landau coefficient α against the frozen oxygen height z_O at $c = 9.9$ Å (vacuum cell, three collinear points: the production $z_O = 0.96$ Å and two dedicated checks at $z_O = 0.90$ and 1.02 Å) has slope $d\alpha/dz_O \approx -0.52$ eV/Å^2 per Å, the same charge-transfer channel that sets Δv, made explicit as a single derivative rather than left implicit in the total-energy scan.

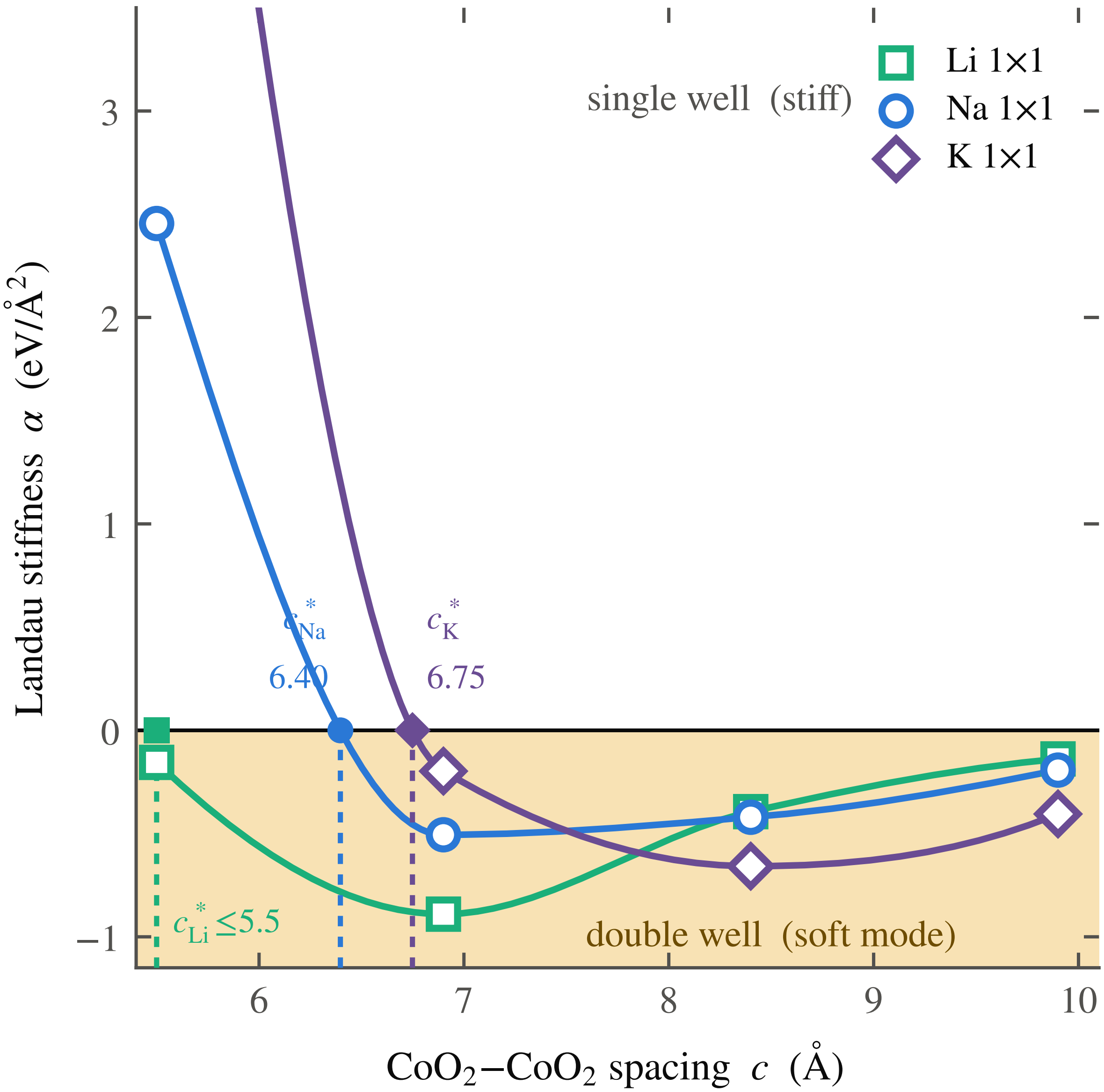

FIG. S10. The threshold ordering, in one plot. $\alpha(c)$ for Li, Na, and K 1×1, with the three thresholds marked: $c^*_{\rm Li} \leq 5.5$ Å (an upper bound only, α is already negative at the smallest spacing sampled, not a bracketed crossing), $c^*_{\rm Na} \approx 6.40$ Å, and $c^*_{\rm K} \approx 6.75$ Å (both bracketed crossings). The threshold rises with ionic size, the prediction behind the hydrated-K_xCoO_2 test proposed in the Letter.

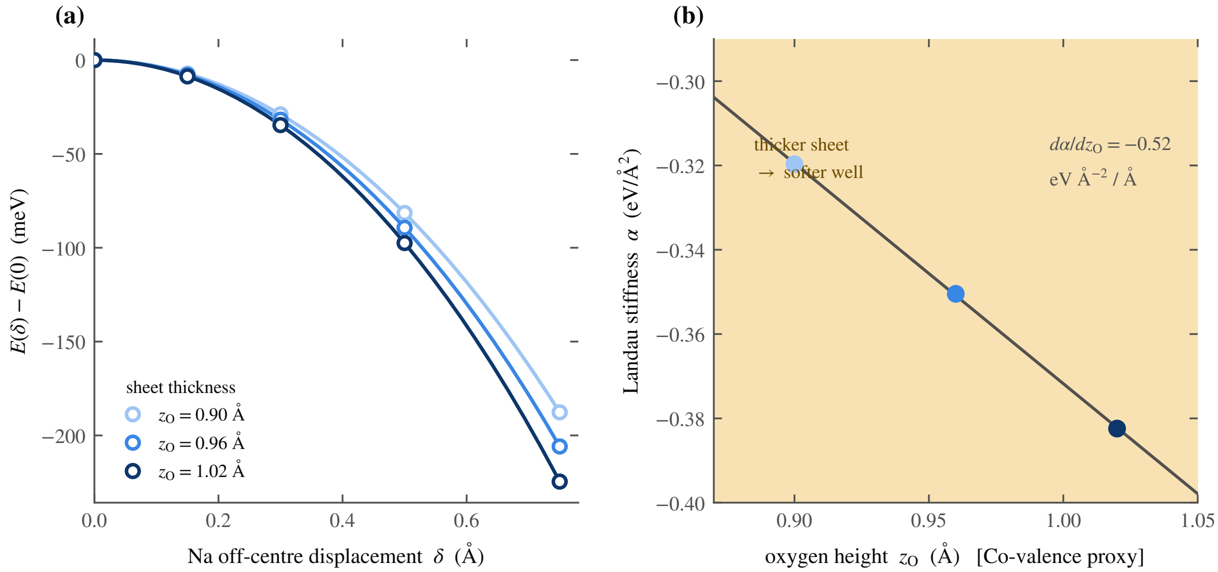

FIG. S11. The microscopic basis of the valence prediction, all vacuum (water-free) potentials at $c = 9.9$ Å. (a) The raw $E(\delta)$ curves at the three sampled oxygen heights $z_O = 0.90$, 0.96 (production), and 1.02 Å, shown for transparency. (b) The extracted Landau coefficient $\alpha(z_O)$: three collinear points give slope $d\alpha/dz_O \approx -0.52$ eV/Å^2 per Å. The defensible quantity is this slope, not an absolute well stiffness; it is the same oxygen-to-gallery charge-transfer channel that sets the valence split $\Delta v \approx 0.06$ per cobalt quoted in the text.

Only the slope is defensible here (three points define a line, not an absolute stiffness); it is offered as the microscopic rate behind Δv, not as a fourth independent calibration of it.

C. Gating and a negative control

A $LiCoO_2$ thin film in which the carrier density is set electrostatically rather than chemically should show the gallery turn-on without the disadvantage of the light frozen ion. Adding carriers to the anhydrous (5.5 Å) lithium cell by a compensating background, 0.1–0.3 e per formula unit, fills the gallery: the Fermi level and the areal density rise monotonically, to $n_{2D} \approx 4 \times 10^{14}$ cm^{-2} at $0.3\,e$, and the lithium well softens as it fills while remaining under a millielectronvolt deep, far below the lithium zero-point energy of ~ 9 meV, so the ion stays dynamically centred. Gating thus tunes the gas without freezing the ion, exactly the place to test the gas and the magnetism apart from the glue, an ionic-liquid-gating prediction for a thin-film collaboration.

Na_2CoSe_2O [31], a mixed-anion cobaltate ($T_c = 6.3$ K) that carries the same two-dimensional cobalt network with a different gallery chemistry, asks whether its gallery ion hosts an interlayer band. It does not: the states at the Fermi level are the Co $3d$–Se $4p$ network, and the ionic sodium of its Na_2O gallery sits well above the Fermi level, with no sodium-derived band there. Na_2CoSe_2O superconducts through its cobalt network, not the gallery, and is a clean negative control: same CoO_2-like sheet, no gallery gas, superconductivity by another route.

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